

IMPLEMENTATION OF A CLOUD PROCESSING BASED VOICE COMMUNICATION AND NOISE REDUCTION EMBEDDED SYSTEM NETWORK

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Abstract:

The use of voice processing systems is widely known and it is a research field that has been explored over the last decades. With the current booming concepts such as Internet of Things and Cloud Processing, there are plenty of options for relying complex processing algorithms to be performed by remote processors. In this research, a new concept is introduced for a remote audio processing network with the use of a small embedded system network. The purpose of the implemented system is to save hardware costs at the expense of cloud processing services by having only low grade 8-bit microcontroller recording units on the end nodes connected to a more powerful small embedded system server that can perform signal processing locally, either with the aid of cloud based processing or the use of both as an optional processing function. The server can act as a local router that can manage different incoming nodes that might perform a previous handshake and device management through TCP/IP common socket protocols. While the connection of the end nodes is wired for implemented test, it is completely possible to turn the system wireless with the use of a WiFi. The end nodes encode the recorded audio to common 16-bit wave encoded file that is uploaded via the router. The router's processing management and interaction is performed in a two stage cloud interaction: the first step consists of utilizing a file management cloud for sound file upload and download while a second script processing cloud interacts with the cloud file management system to retrieve the incoming files and perform noise reduction techniques for proper file download and output by the requesting node. In order to record and play the recorded sounds, the use of a simple microphone and speaker/earphones were used. The entire node to node communication did not exceed 2 minutes for a half-duplex communication. The intention of implementing this kind of system was to design a low cost inter-phone/broadcast system in a local area that could save the cost of a high end processor in the end node size by means of useful software libraries and tools that are currently available in the cloud. This system could also have the capability to send the recorded files to a cloud service in order to classify the input as a specific command in order to control remote devices by means of an Internet of Things system. If higher end processors such as the used for the router were used as end nodes, it could be possible to establish a full-duplex network with the aid of cloud noise smoothing functionality.

Keywords: Cloud Computing, Embedded system network, Internet of Things, Sound processing

1. BRIEF CLOUD COMPUTING AND INTERNET OF THINGS BACKGROUND

The concept of Internet of Things (IoT) is a very recent trend. Moreover, the implementation of embedded systems with IoT capabilities is increasing in the years to come. The concept relates the internetworking of physical devices, vehicles or smart devices through Internet [1]. Most of the current ways of communication for IoT platforms include the use of LAN, Wi-Fi, 3G or LTE technologies; protocols such as RFID, BLE or QR also include the capacity for allowing devices to send information through Internet [1].

The importance of connecting devices through the network also includes the possibility of analyzing and synthesizing information for remote monitoring and processing purposes. The concept of cloud computing includes the use of a distant server that receives a stream of data in order to process or display information through a graphical user interface or webpage. The use of this type of architecture allows saving infrastructure costs since tasks are executed remotely by a service provider. Cloud computing is a new terminology that was added to IT jargon in early 2007[2]. Cloud computing as a business and technical model derives many of its benefits from other terminologies such as distributed computing. The advantages include better hardware utilization (and component reduction, as with this research purpose), higher revenues, bigger software markets, as well as higher system scalability. On the other side, dependability on the network and availability might constitute some turndowns for a platform migration.

In this research, a small embedded system network was configured to test voice recording and processing features in a way that hardware costs could be affordable in exchange for a reliable cloud processing service. The use of common sound smoothing techniques and device design is introduced as well as the whole system development is presented.

2. NOISE REDUCTION ALGORITHMS

In order to assign a signal processing task for the cloud server, two different algorithms were analyzed in order to perform simple signal smoothing. The two approaches are as follows.

The Simple Moving Average Model (SMA)

By defining a window size n , an unweight mean is obtained for a vector $x(n)$ containing a series of data (Equation 2.1). The mean is normally taken from an equal number of data on either side of a central value [3].

$$SMA = \frac{1}{n} \sum_{i=0}^{n-1} x_{n-i} \quad (2.1)$$

SMA has important applications in Economy. However, in this research, the smoothing of the collected files allowed to have a quality analysis against other noise smoothing techniques. There are different moving average calculation methods, the simple option was chosen for this paper as described later.

Median Filter

A popular alternative to averaging, which is sure to produce a better estimate when the average is taken over a homogeneous neighborhood without noise, is the median filter that replaces the value the median value of the neighborhood.

The median filter is a nonlinear digital filtering technique, often used to remove salt and pepper noise. Let $A[i]_{i=0 \dots (n-1)}$ be a sorted array of n real numbers. The median of the set of numbers in A is $A[(n-1)/2]$. Median filtering is very widely used for boundary preservation. Similarly to Moving Average, there are very different options for its calculation and performance based on window size. However, for this research, a Mean Median Filter (MMF), consisting of an odd length fixed sized scanning window [4] was studied to analyze its effects and compare it to the previous methodology.

3. CLOUD SERVICE SYSTEM DESCRIPTION

The idea of having a voice processing system relying on a cloud based service might represent a problem in terms of delay. However, if the purpose of the system is to provide communication service for a short time, it might become a feasible idea. Currently, voice systems rely on the use of high speed voice embedded devices on the intercom node, while connecting the devices via a wired or wireless method. In this case, if the number of hardware components on the end nodes could be reduced to a lowest, the use of a fast and reliable remote cloud service could represent a long term savings in terms of manufacturing costs: also an open gateway for conversation logging and implementation of remote device operation with the use of this methodology by remote speech recognition algorithm execution. The proposed topology that was tested in this research is shown in Figure 1

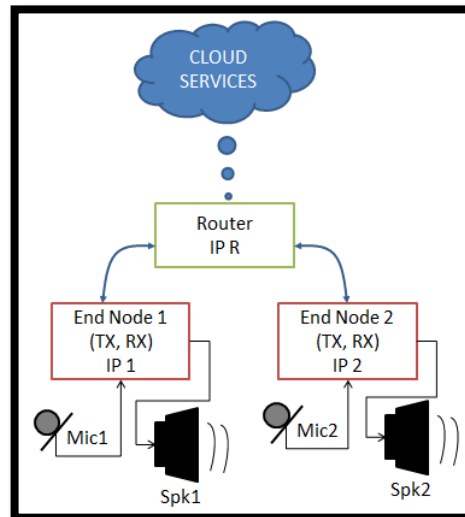


Figure 1. Figure caption sample

The cloud services consisted of a two layer cloud service system (Figure 2). The first layer was in charge of interacting directly with the router through remote file exchange with the use of a known file management cloud with file upload, edit and download capabilities via a

web based API [5]. In order to access to a particular remote file system, an access token was required through web APP generation. There are many different ways to access a restricted cloud storage space with the use of a user generated token, which can be obtained through verification or by providing it to the local router system. The main cloud layer is in charge of retrieving the file cloud sound to be processed and perform the noise reduction. The used cloud service [6] was capable of processing the files on either a preprogrammed task scheduling routine or running the processing script live. However, the latter is not a cost effective solution due to the fact of running a process at all times by consuming the remote equipment's available processing time.

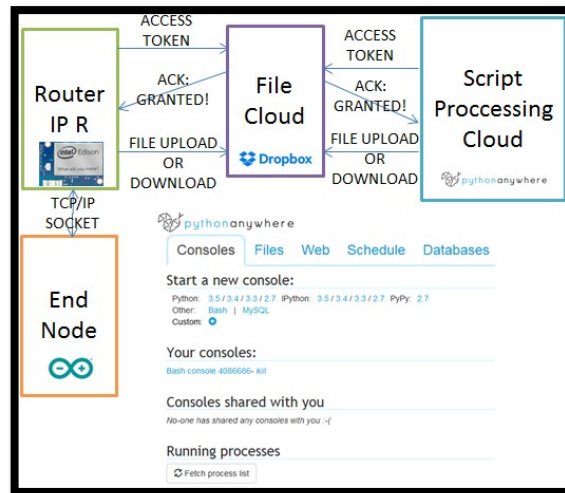


Figure 2. Description of the cloud service

4. EMBEDDED SYSTEM NETWORK IMPLEMENTATION

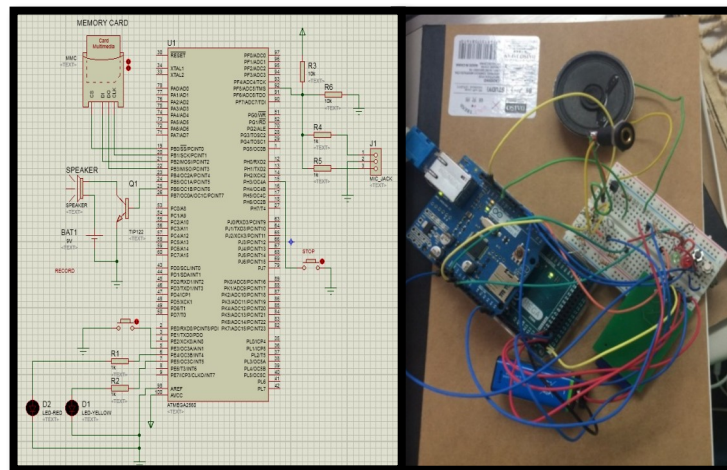


Figure 3. End Node Design and Implementation

The end node design consisted of a mix of recorder and player hardware and software implementation. The software digital voice recorder implementation required a simple ADC system with the use of simple resistors for signal sampling and an SD card for saving writing the compressed data into wave format (WAV) [7][8]. Although audio quality is not expected to be very good, smoothing techniques could enhance the signal as explained later. In order

to play the received messages, a small hardware consisting of a low power stage with a switching transistor, a conventional Buffer and battery was implemented. WAV files from an SD card can be played with the aid of sophisticated software libraries [9]. Figure 3 shows a hardware diagram and its real implementation. The embedded hardware uses an Arduino Mega 2560 due to the microcontroller's high memory capacity and also the use of an Ethernet layer for Network communication. The router consisted of a microprocessor based device with WiFi capability, an Intel Edison board (Figure 4) [10]. The router is able to accept connection request from different end nodes and classify the communication type (upload or download) within the established network. Also, it is in charge of communicating with the file transfer cloud layer. The common transfer protocol (due to the WAV transfer requirements) was through common TCP/IP sockets. The data chunks for uploading the recorded voice WAV files to the router was 4 KB (4KB per transaction operation) while the downstream was radically reduced to 1 byte per second due to the Ethernet layer low downstream capabilities (that could be fixed with proper software implementation on later stages). The IoT router allows a fast communication with end nodes and the file transfer cloud service layer.



Figure 4. Embedded System Network Router device

5. Implementation Results

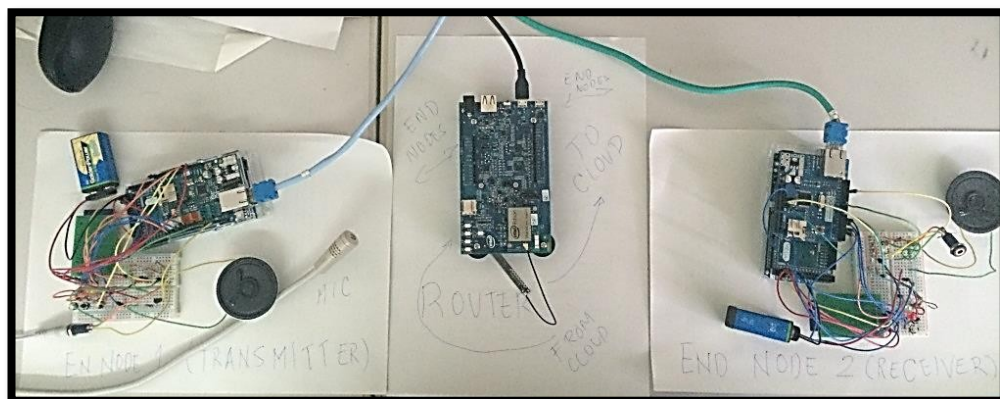


Figure 5. Embedded System Network Implementation

In order to realize one way communication between two nodes, a two end-node and router system was implemented as shown on Figure 5. For the experiment, one side contained a microphone that could perform signal recording and processing with the cloud script

execution service. As presented on Figure 6, the smoothing performed by median filter and moving average (with a 10 sample size window) is compared for a test recorded file. The sampling rate of the wave files is 22,000 Hz for a 16-bit encoding format, with a total length of about 5 seconds. In terms of quality, the signal processed by moving average provided a better audio quality for listeners in comparison to the median filter, probably due to the fact that the recorded signal error was relative to the sampling error trend only at moist sampling points and the recorded file did not contain salt and pepper noise on its whole.

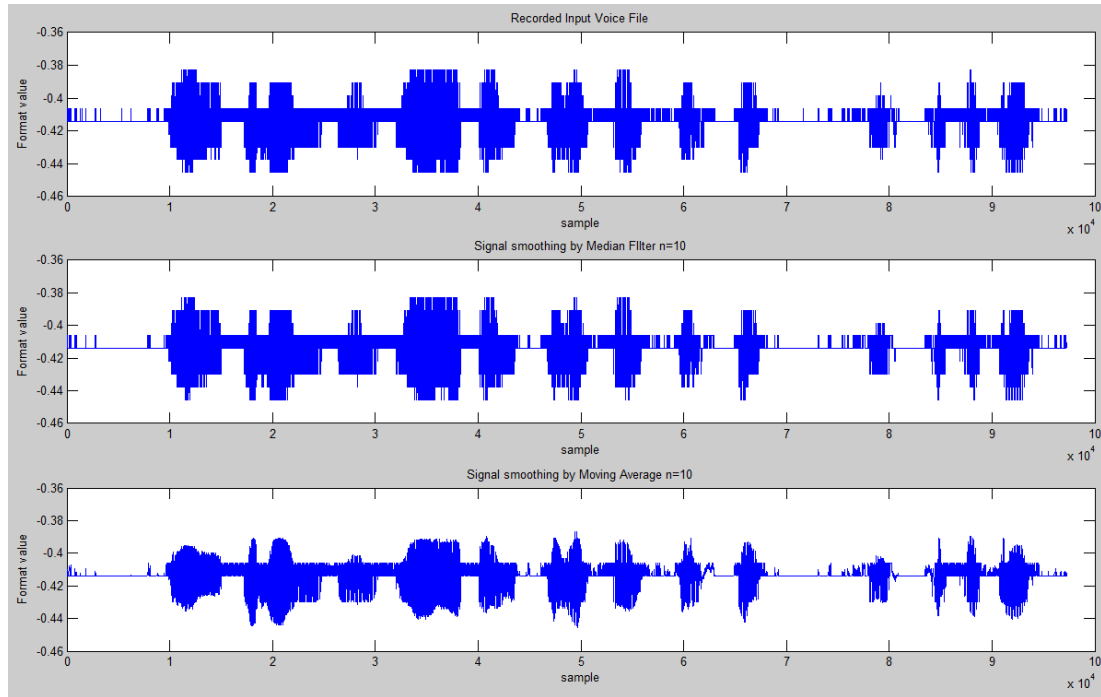


Figure 6. Original test signal(Top), Median Filter(Middle) and Moving Average(Bottom) Smoothing

The total length of the audio file was 95.0 KB and the entire time of transfer and processing between nodes was no longer than two minutes. Currently, the system has a low downstream throughput due to the Ethernet stack low download software library performance. However, good audio communication between node ends is possible in a half-duplex communication within a local network with the aid of a remote service.

6. Conclusions and Further Work

In this research end low end hardware audio recording device transfer and processing with the use of a remote service could be achieved in a half bridge configuration. The delay time between end nodes is reduced to solving the low downstream performance by implementing the software function on the download node's side Ethernet stack. However, the signal smoothing provided a good output quality. The power stage of the end speakers could also be improved with the use of a different voltage level or adjusting by means of low end hardware arrangement. The cloud service layers proved to be efficient.

The application of the current configuration can be applied to local communication systems that require one way communication like remote speakers, short call interphones; the use of a local router can also allow the signal communication. Also further work could include the use of remote speech recognition and classification for remote command performance.

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